# ATEK303 cheat sheet

ACID FIRMWARE FOR THE M-VAVE FM-1 · BY ANIMATEK

**PRESS PLAY**

a line plays · turn the knobs · it moves

v1.1 · GPL-3.0 · github.com/animatek/FM1-Animatek

## QUICK START

1. `PLAY` starts the pattern.
2. `K1` – `K4` cutoff, reso, env mod, decay.
3. `SEQ` a brand-new acid line.
4. `ENV` `LFO` `ARP` mutate it; `REC` undoes.
5. Press a key: the line transposes, like on a 303.
6. `SAVE` twice keeps it as a preset with an acid name.

## BUTTONS

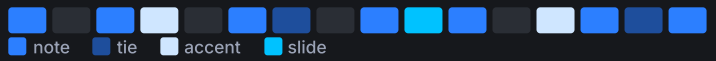
| | |
|---|---|
| `PLAY` | run / stop (internal clock, or MIDI clock) |
| `SEQ` | GENERATE a new pattern (new seed) · playing: at the next bar |
| `ENV` | mutate TIME (rests, notes, ties) · on the page in view |
| `LFO` | mutate NOTES (octaves) |
| `ARP` | mutate SLIDES and ACCENTS |
| `REC` | undo the last mutation or edit |
| `GLO` | LOCK: SEQ mutates all instead of a new seed |
| `EDIT` | step editor on / off |
| `HOME` | VCF → VOICE → PATT → SEQ; from anywhere else, VCF |
| `OCT-` + `HOME` | OPTIONS: colour, MIDI channels, seq notes / clock out, fine tuning |
| `FX` | effect pages (again: the next one) |
| `SAVE` | save screen: slot, name, OVERWRITE? NO / YES |
| `OCT-` `OCT+` | octave (OCT– on release) |
| keys | stopped: play the voice (hard = accent) · running: transpose |

## 8 PAGES · SELECT · K1–K4

| | | | |
|---|---|---|---|
| **VCF** · VOICE 1 | CUTOFF | RESO | ENV MOD | DECAY |
| **VOICE** · VOICE 2 | ACCENT | TUNE | WAVE saw/sqr | GATE length |
| **PATT** · PATTERN 1 | DENSITY | RANGE | ACCENTS | SLIDES |
| **SEQ** · PATTERN 2 | STEPS 1–64 | SCALE | TEMPO | ROOT |
| **DELAY** · FX 1 | TIME 1/16–1/4. | FEEDBACK | TONE | MIX |
| **CHORUS** · FX 2 | TYPE chor/flang | RATE | DEPTH | MIX |
| **DRIVE** · FX 3 | TYPE od/rat/fuzz | DRIVE | TONE | LEVEL |
| **REVERB** · FX 4 | DECAY | DAMP | PRE-DELAY | MIX |

DENSITY, RANGE, ACCENTS and SLIDES regrow the pattern from its seed in real time. SCALE and ROOT change at the output: the line keeps its shape. Up to 64 steps: 4 pages of 16, the page playing on view. Signal: voice → drive → chorus → ping-pong delay (synced) → plate reverb.

## STEP EDITOR · EDIT



note · tie · accent · slide

| | |
|---|---|
| `SELECT` · `K4` | step (the 16 of the page) · page 1:4 … 4:4 |
| a key | sets the step's note and moves on |
| `K1` `K2` `K3` | gate (REST, NOTE, TIE) · octave (UP/DOWN) · semitone |
| `OCT-` + `K4` | hold OCT– and turn K4: STEPS (1–64) |
| `ENV` `LFO` | toggle ACCENT · toggle SLIDE |
| `REC` · `EDIT` / `HOME` | undo · leave the editor |

## PRESETS · 32 SLOTS

`SAVE` opens the save screen: `PRESETS` picks the slot, `SELECT` a new acid name, `SAVE` stores everything (pattern, voice, effects, tempo, scale, root); over a used slot it asks OVERWRITE? NO / YES. Any other button cancels.

12 factory presets in slots 1–12; a new FM-1 starts on 01. Turn `PRESETS`: the list opens and loads each preset as the cursor reaches it; while playing it waits for the end of the pattern, like a 303 (**NEXT**). `ENV` twice deletes, `SAVE` saves on that slot, `HOME` closes (or 6 s untouched).

## MIDI

In: stopped, notes play the voice; running, they transpose. Clock, start, stop and continue drive the sequencer. Out (USB): the keys and the sequencer's notes (accent 127); clock out on request. Channels in `OPTIONS`.

| | | | |
|---|---|---|---|
| 74 cutoff | 71 reso | 75 env mod | 76 decay |
| 77 accent | 78 tune | 79 wave | USB audio to PC |

## INSTALL · LEAVE · RESCUE

Install from The Firmware Hub (Chrome or Edge), charged battery, cable straight in. It carries the Felucca update loader: any installer can put another firmware or M-VAVE's back.

Fails to start twice? It drops to a safe mode (PLAY retries, REC updates). `REC` held at power-on: the chip's update mode.